Emerging Methods of Auction Design in Social Networks

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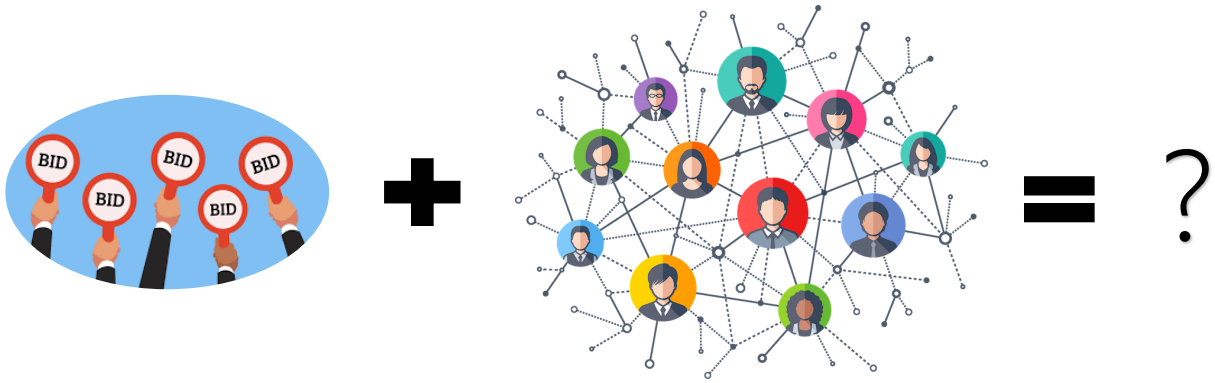
Introduction of Networked Auction

Backgrounds

- **Auction**: A paradigm for reasonable resource allocation;
- **Traditional Vickrey Auction**:
  - Spreads the sale information;
  - Bidders bid for the item;
  - Decide allocation and payment.
- **Networked Diffusion Auction**:
  - Every participant can invite their friends;
  - All bidders bid for the item;
  - Decide allocation and payment.



Introduction of Networked Auction

Benefits and Challenges

- **Benefits**:
  - Spreading sale information by friends. $\Rightarrow$ no extra cost.
  - Potential higher bid players $\Rightarrow$ higher social welfare and revenue.
- **Challenges**:
  - Conflicts between auctioneer (more players brings payoff) and bidders (tougher competition).
  - No incentive to Diffuse!

Introduction of Networked Auctions

General Models

- A set of k (identical/non-identical) items for sale.
- One (weighted/unweighted) network $G = (N, E)$. Nodes set N: bidders set; Edges set E: interaction between bidders.
- Bidder i's type $t_i = (v_i, r_i)$. v_i: valuation function (unit-demand/additive/downward-sloping/...); r_i: diffusion agents set.

Introduction of Networked Auctions

Auction Properties

- Diffusion Mechanism $\mathcal{M} = (\boldsymbol{\pi}, \boldsymbol{x})$.
  $\boldsymbol{\pi}$: allocation policies; $\boldsymbol{x}$: payment policies.
- Bidder i's utility: $u_i(\boldsymbol{t}) = \pi_i \cdot v_i - x_i$
- Individual Rational (IR): truthful report v_i: $u_i \geq 0$.
- Incentive Compatible (IC): truthful report t_i (not just v_i): $u_i(t_i, \boldsymbol{t}'_{-i}) \geq u_i(t'_i, \boldsymbol{t}''_{-i})$.
- Efficient: maximizing social welfare.
- Weakly Budget Balance (WBB): No deficit for the seller.
- There exists trade-off between efficiency, IC and budget balance.

Networked Auctions	Single-Item Auctions	Unweighted Graph	Diffusion-VCG [Li *et al*. 2017]
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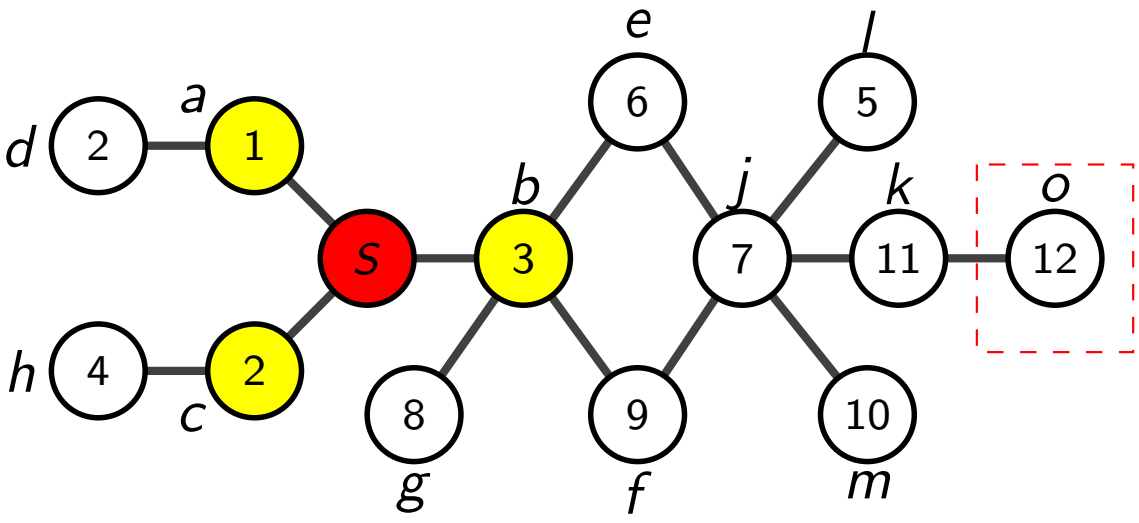
A Starter: Information Diffusion Mechanism

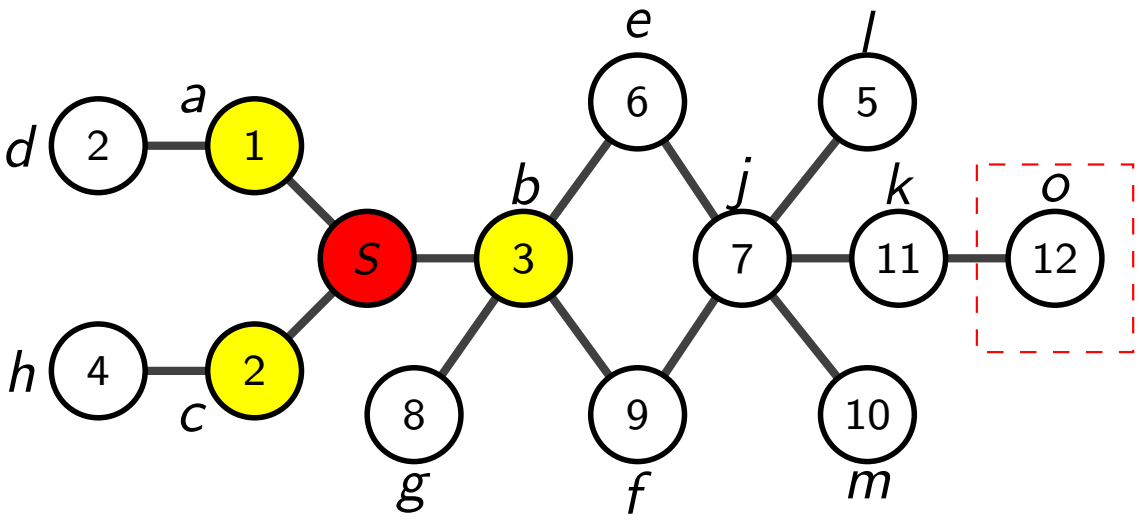
- **Allocation**:
  - Critical diffusion sequence of highest bid.
  - First bidder: $v_i = v^*_{-(i+1)}$.
- **Payment**:
  - Winner: v^*_{-i}
  - Cut nodes on winner's sequence: $v^*_{-i} - v^*_{-(i+1)}$.

A Starter: Information Diffusion Mechanism



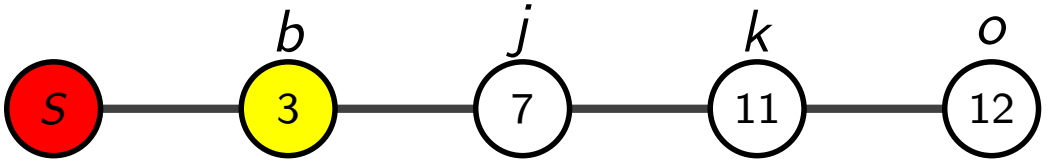
Step.1 Finding the agent with highest bid: $o(12)$

A Starter: Information Diffusion Mechanism

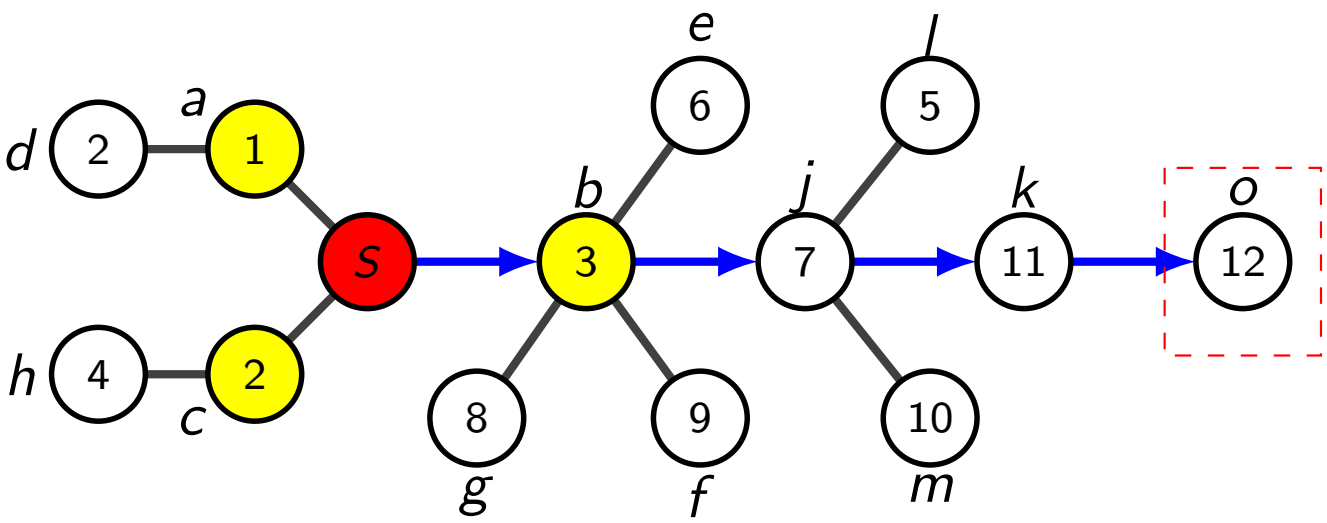


Step.1 Finding the agent with highest bid: $o(12)$

Step.2 Create critical diffusion sequence of o:



A Starter: Information Diffusion Mechanism



Bidder	v_i	v^*_{-i}	$v^*_{-(i+1)}$	π_i	x_i	u_i
b	3	4	9	0	$4-9=-5$	5
j	7	9	10	0	$9-10=-1$	1
k	11	10	11	1	10	1
o	12	11	*nil*	0	0	0

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IR & IC Diffusion Auctions

Framework

IR and IC Diffusion Auctions [1]:

- ✓ $\boldsymbol{\pi}$ is value-monotonic.
- ✓ $\boldsymbol{x} = \boldsymbol{\pi}\tilde{\boldsymbol{x}} + (1 - \boldsymbol{\pi})\bar{\boldsymbol{x}}$. $\tilde{\boldsymbol{x}}$:winning payment; $\bar{\boldsymbol{x}}$:losing payment
  $\tilde{\boldsymbol{x}}$ and $\bar{\boldsymbol{x}}$: bid-independent and diffusion monotonic.
- ✓ For bidder i: $\tilde{x}(r_i)$ - $\bar{x}(r_i) =$ (critical bid) $v^*(r_i)$.
- ✓ For bidder i: $\bar{x}(\emptyset) \leq 0$.

An extended version of Myerson's Lemma in diffusion auctions!

[1] Bin Li, Dong Hao, Dengji Zhao: Incentive-Compatible Diffusion Auctions. IJCAI 2020.

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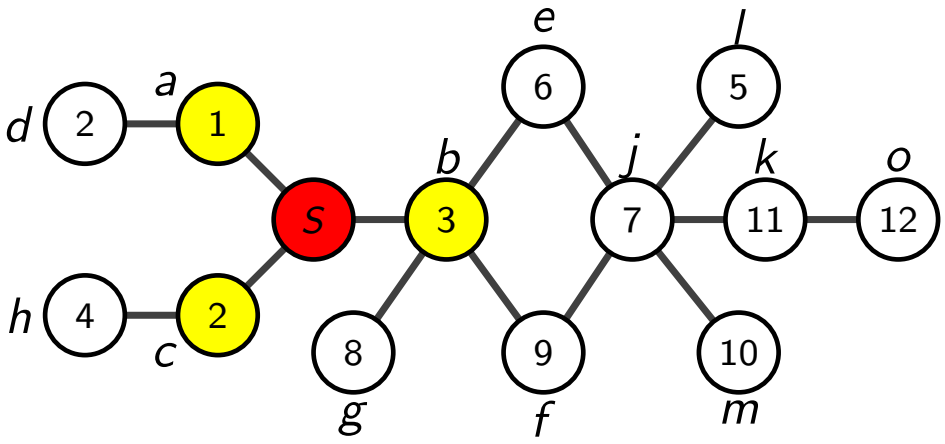
Single-Item Networked Auction via Unweighted Graph

Examples



$\mathcal{M}$	Winner	Rewarded bidders	SW	$Rev^{\mathcal{M}}$	Efficient
SPA	$b(2)$	$\emptyset$	3	2	✗
VCG	$o(11)$	$b(-8), j(-3), k(-2)$	12	-2	✓
IDM	$k(10)$	$b(-5), j(-1)$	11	4	✗
CDM[1]	$k(10)$	$b(-4), j(-1)$	11	5	✗
FDM	$k(10)$	$b(-4), e(-\frac{1}{3}), j(-1)$	11	$\frac{14}{3}$	✗
NRM	$k(10)$	$a(-\frac{8}{13}), b(-\frac{18}{13}), c(-\frac{4}{13})$,$e(-\frac{5}{8})$ $f(-\frac{1}{2}), g(-\frac{5}{8}), j(-\frac{5}{2}), l(-\frac{1}{4})$	11	$\frac{61}{26}$	✗

Table 1: Numerical results of some representative diffusion mechanisms

[1] Here take α_i as the minimum cutting edges subset blocking next critical diffusion node.

Single-Item Networked Auction via Unweighted Graph

Examples

$\mathcal{M}$	Winner	$\lvert Rewarded\ bidders \rvert$	SW	Efficient	$Rev^{\mathcal{M}}$
SPA	b	0	3	✗	2
VCG	o	3	12	✓	-2
IDM	k	2	11	✗	4
CDM	k	2	11	✗	5
FDM	k	3	11	✗	$\frac{14}{3}$
NRM	k	8	11	✗	$\frac{61}{26}$

- Vickrey Auction without diffusion: low SW and $Rev^{\mathcal{M}}$.
- Diffusion-VCG[1]: Efficient but not WBB.
- IDM[1]: The very first work; seen as one special case in CDM. CDM[2]: One diffusion mechanisms family; min-cutting edges set: highest $Rev^{\mathcal{M}}$ in this example.
- FDM[3],NRM[4]: Give rewards for more diffusion nodes.

[1] Bin Li, Dong Hao, Dengji Zhao, and Tao Zhou. Mechanism design in social networks. AAAI 2017

[2] Bin Li, Dong Hao, Dengji Zhao, and Makoto Yokoo. Diffusion and auction on graphs. IJCAI 2019

[3] Wen Zhang, Dengji Zhao, and Yao Zhang. Incentivize diffusion with fair rewards. ECAI 2020

[4] Wen Zhang, Dengji Zhao, and Hanyu Chen. Redistribution mechanism on networks. AAMAS 2020

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Single-Item Networked Auction via Weighted Graph

- Allocation rule needs to consider transferring cost.
- Non-critical diffusion nodes change the allocation results!

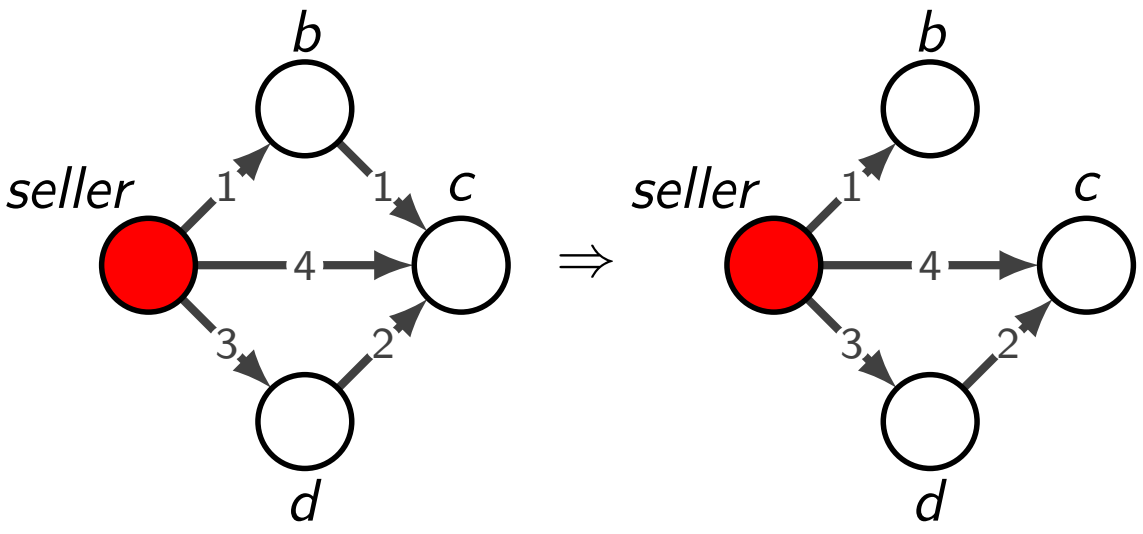

Figure 1: Small Part of One Weighted Graph

b not diffuse to *c*, trading cost changed!

- Critical diffusion sequence does not work!

Single-Item Networked Auction via Weighted Graph

- WDM[1] in General weighted graphs:
  - **Allocation**: First bidder satisfying $v_i = v^*_{-\gamma_i}$ in min-cost trading path.
  - **Payment**:
    Winner:Critical bid beating all on-path nodes before her.
    Bidders on path: Social welfare decrease correlated with their cutting setγ_i.
- CSM[2] in Graphs with intermediaries:
  - **Allocation**: The agent whose trading path would maximize the social welfare.
  - **Payment**:
    Winner: VCG payment.
    Intermediaries: Social welfare decrease correlated with their threshold neighbourhood set.

Idea: Encouraging Sharing customers among merchants.

[1] Bin Li, Dong Hao, Dengji Zhao, and Makoto Yokoo. Diffusion and auction on graphs. IJCAI 2019

[2] Bin Li, Dong Hao, Dengji Zhao, and Tao Zhou. Customer sharing in economic networks with costs. IJCAI 2018

Research Map of Diffusion Auction Mechanisms

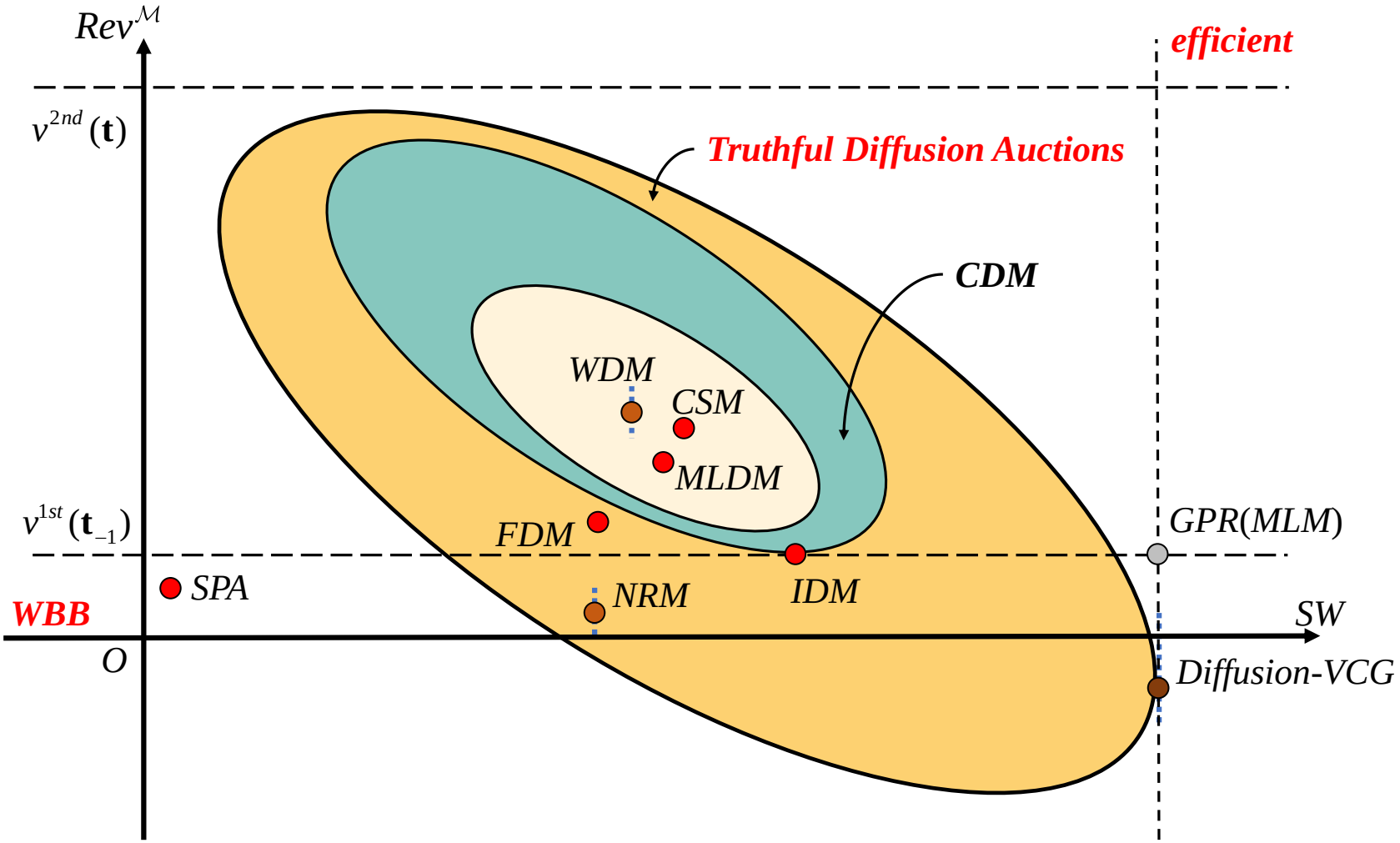

Figure 2: Research Map of Single-Item Diffusion Auctions

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Multi-Unit Networked Auction

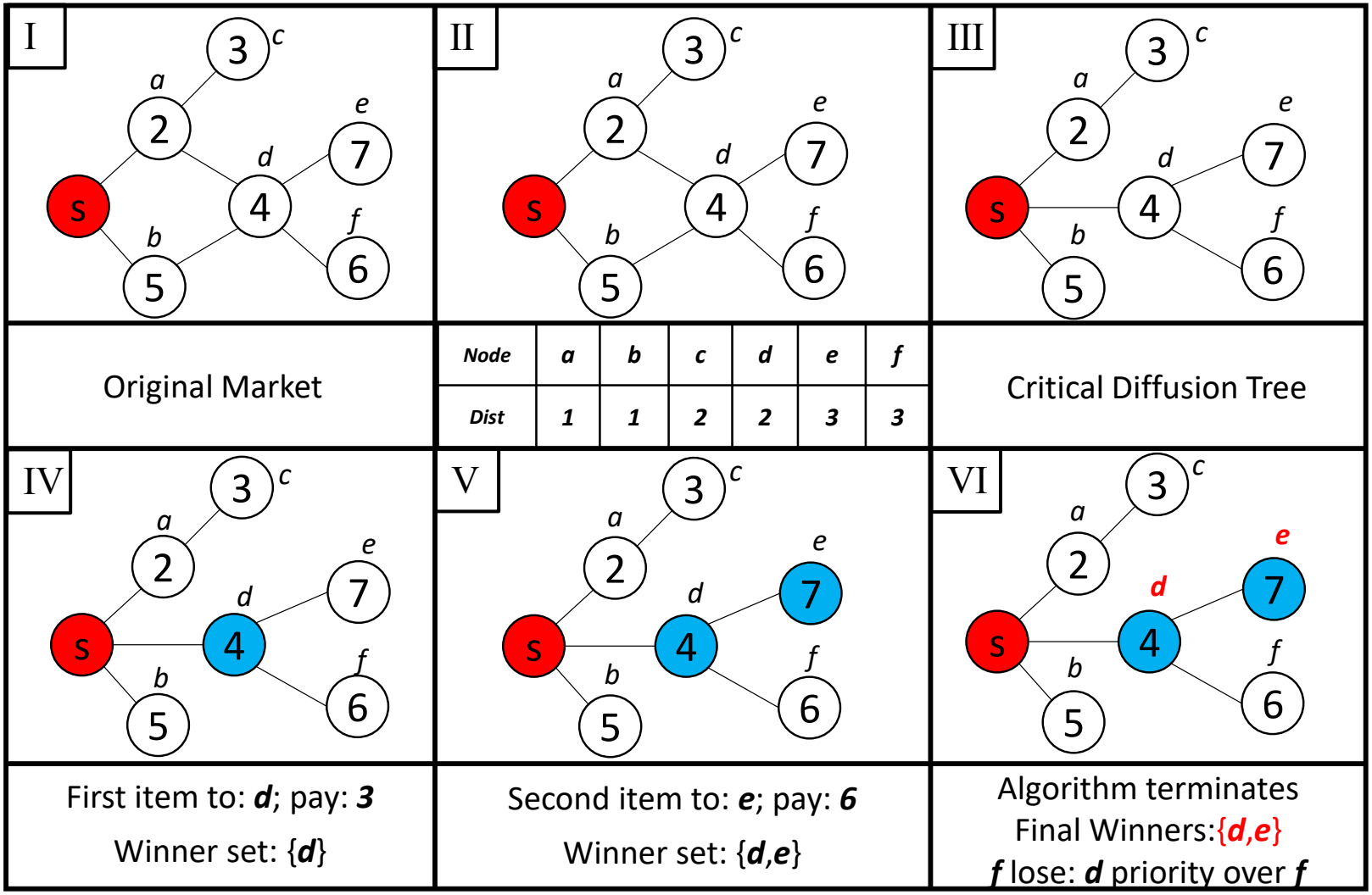
Unit-demand bidders

Challenges:

- Misreporting $\Rightarrow$ change allocation of goods to different branches.
- Bidders could change their ancestors' and children's allocation & payment. $\Rightarrow$ gain more payoff.
- How to restrict agents' action space for truthfulness?

Multi-Unit Networked Auction

One Solution: Distance-Based Networked Auction(DNA-MU)[1]



[1] Takehiro Kawasaki, Nathanael Barrot, Seiji Takanashi, Taiki Todo, Makoto Yokoo, et al. Strategy-proof and non-wasteful multi-unit auction via social network. AAAI 2020

Multi-Items Networked Auction

Some Other Scenarios

- GIDM [1]: Multi-unit networked auction with unit-demand bidders.
- GAPM [2]: Multi-unit neworked auction with down-sloping valuation.
- DNAM[3]: Networked auction in independent double auction markets.
- BDM-M/H[4]: Networked procurement auction with budget.

[1] Dengji Zhao, Bin Li, Junping Xu, Dong Hao, and Nicholas R. Jennings. Selling multiple items via social networks. AAMAS 2018

[2] Seiji Takanashi, Takehiro Kawasaki, Taiki Todo, and Makoto Yokoo. Efficiency in truthful auctions via a social network. arXiv preprint arXiv:1904.12422, 2019

[3] Junping Xu, Xin He, and Dengji Zhao. Double auction design on networks. DAI 2019

[4] Liu, Xiang, Weiwei Wu, Minming Li, and Wanyuan Wang. Budget Feasible Mechanisms Over Graphs. AAAI 2021

Diffusion Incentive Design beyond Auctions

- Crowdsouring: DARPA 2009 red balloon challenge[1]; data acquisition via social networks[2];
- Matching[3]: Modified TTC under acyclic preference and tree network.
- Information Propagation with budget limit[4].
- Influence Maximization in social networks[5].

Different from diffusion auctions: agents' type space, target$\cdots$

[1] Galen Pickard, Wei Pan, Iyad Rahwan,Manuel Cebrian, Riley Crane, Anmol Madan, and Alex Pentland. Time-critical social mobilization. Science 2011

[2] Wen Zhang, Yao Zhang, and Dengji Zhao. Collaborative data acquisition. AAMAS 2020

[3] Takehiro Kawasaki, Ryoji Wada, Taiki Todo, and Makoto Yokoo. Mechanism design for housing markets over social networks. AAMAS 2021

[4] Haomin Shi, Yao Zhang, Zilin Si, LetongWang, and Dengji Zhao. Maximal information propagation with budgets. ECAI 2020

[5] David Kempe, Jon Kleinberg, and Eva Tardos. Maximizing the spread of influence through a social network. KDD 2013

Open Questions in Diffusion Mechanism Design

- **New Models of Diffusion Auction**:
  - ✏ Optimal networked auction design.
  - ✏ Networked combinatorial auctions?
- **Properties of Diffusion Auction**:
  - ✏ Robustness of diffusion auctions: sybil-proofness and collusion-proofness?
  - ✏ Different networks influence result of mechanisms?
- **Other Diffusion Incentive Design**:
  - ✏ Voting: Reducing cost under networked voting while guaranteeing efficiency?
  - ✏ Rating: Extending personalised rating systems with more agents?
  - ✏ Any other scenarios need attracting more participants?

Summary

Networked Diffusion Auction:

- ✓ **Market Expansion** by participants' invitation.
- ✓ Real-life apps: Pinduoduo, Groupon shopping platform, $\cdots$

Figure 3: PinDuoDuo

Figure 4: Groupon

- ✓ **Promising direction for better optimizations**:
  More reasonable resource allocation: increasing efficiency, revenue, $\cdots$

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